Fairness to Footprint: A Comprehensive Audit Framework for Sustainable AI Systems

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Abstract

The growing use of intelligent systems across different industries has brought increased attention to issues like fairness, transparency, and environmental impact when deploying these technologies. This work introduces a comprehensive auditing method that evaluates such systems from ethical, legal, and ecological perspectives. This approach combines bias detection tools like FairLearn and AIF360, explanation techniques such as LIME and SHAP, and carbon footprint monitoring with CodeCarbon to offer an accessible web-based platform for uploading and evaluating models and datasets. It includes built-in checks to ensure adherence to important privacy laws, including HIPAA and GDPR, helping users stay compliant with regulatory requirements. Detailed audit reports are generated, featuring visualizations and performance summaries that guide users in identifying risks and taking corrective measures. Previous research using similar methods has successfully cut bias by up to 54% and improved model transparency by 40%. It has also significantly decreased related carbon emissions by 20 to 30%. Its modular design allows it to be adapted for different models, datasets, and legal contexts. This supports the ethical, transparent, and environmentally responsible use of new technologies worldwide.

Keywords: AI Auditing, Fairness Assessment, Bias Detection, Explainable AI, SHAP, LIME, Energy Efficiency, Carbon Footprint, CodeCarbon, Ethical Compliance, GDPR, HIPAA, Automated Reporting, Model Interpretability, Modular Pipeline, Accountability, Transparency, Sustainable AI

1. Introduction

The extensive adoption of intelligent systems has transformed decision-making and workflows across industries such as healthcare, finance, and public administration. These systems have often improved efficiency and brought challenges related to environmental impact, fairness, legal adherence, and transparency. Since AI models influence critical decision-making, challenges related to bias, transparency, and ethical use have become key in global research [4].

In recent years, researchers have developed a variety of frameworks and tools that extend beyond conventional accuracy-based evaluations [6]. Recent progress involves developing adaptable mechanisms to evaluate across bias within data and algorithms. Fairness toolkit like Fairlearn, combining SHAP and LIME techniques, which play a vital role in clearly understanding model predictions.

Awareness of AI's environmental consequences, as measured by energy use, has encouraged the development of monitoring initiatives like CodeCarbon.

Regulatory bodies have begun to focus on strengthening compliance and governance. Regulations such as GDPR and HIPAA outline maintaining data privacy and responsibility. Such regulations require proving that their AI systems align with global standards. Consequently, there is a growing demand for comprehensive audit solutions that address bias detection, explainability, and compliance [1].

A robust auditing system features a web-based interface for uploading models and datasets. It brings together modular tools for bias detection, interpretability assessment, energy tracking, and compliance validation. It generates detailed visual reports and practical insights. These frameworks link how AI works with how it's managed, helping organizations build and use AI systems people can trust.

This paper reviews existing work on AI auditing, describes how auditing approaches and evaluation measures have evolved, and presents best practices for developing reliable audit workflows [3]. This review highlights both the progress and ongoing difficulties in designing AI systems that are audit-ready, compliant with regulations, and eco-friendly.

2. Literature Survey

The implemented study incorporated multiple open-source libraries to evaluate fairness, explainability, and sustainability in artificial intelligence systems. Each project module focused on a distinct research objective and corresponding evaluation outcome.

- **Transparent and Accurate Energy Forecasting Framework**
  The presented HW+XGBoost+XAI framework combines Holt–Winters Exponential Smoothing with XGBoost and explainable AI technology for multi-scale forecasting of energy consumption. The model accurately predicts energy consumption at time scales of hourly to quarterly levels. Results show the model delivered an R^2 of 0.9999 at the hourly level and 0.9986 at the quarterly level, and SHAP provides global interpretability, and LIME enables interpretability at the local level. Taken together, the proposed methodology improves the accuracy of forecasting consumption and the transparency of the model's predictions. Alternatively, it supports evidence-based decision processes in smart energy management systems [29].

- **Mitigating Bias in AI Predictive models for primary health care**
  AI prediction models in primary health care have the potential to improve the health of populations; however, they can also possess the potential to introduce and exacerbate bias towards groups that are diverse. This article provides a review of the current methods to measure and reduce bias in AI models, particularly with respect to protected attributes, such as race, sex, age and socioeconomic status. Methods assessed include a range of pre-processing approaches, relabeling and reweighing, use of electronic health records, human-in-the-loop interventions, and fairness metrics. Outcomes propose that bias can have the most success in pre-processing approaches, which includes the engagement of participants. Trade-offs to consider were noted in some studies, where reducing bias and proving fairness came with a parallel decrease in overall model performance [36].

- **Energy-Efficient Optimization Framework for AI Model Training and Inference in Enterprise Systems**
  The framework described here provides energy-efficient optimization strategies for training and inferring AI models within enterprise systems. The system uses multi-objective algorithms to decrease energy consumption, but also achieve a high performing model - realizing a reduction in energy consumption of 30.6% with only 0.7% degradation in accuracy. A controlled experimental data center was established by isolating a portion of a large operational data center, with separate cooling systems, environmental sensors, and power management for advancing the validated optimization strategies. A total of 25 runs were observed for comparisons against the baseline system, where all experimental observations showed energy savings during the controlled test. The optimization framework also preserved solid model performance where acceleration degradation averaged only 0.8%. Hence, the framework grouping was effective in maintaining energy advantages and at the same time, maintained functioning capability. Overall, the system establishes probable capability for deploying AI models

sustainably, striking a balance between both energy efficiency and functioning capability in the enterprise domain [30].

- **Algorithmic Fairness framework for Credit Scoring using Bias Mitigation Techniques.**
  This framework demonstrates the use of algorithmic decision-making strategies that facilitate fairness in credit scoring through machine learning models. Evaluated twelve possible bias mitigating strategies on five fairness metrics across two datasets (the German Credit and Consumer Loan datasets), and gauge trade-offs between bias capture, accuracy, and profitability using AI Fairness 360. Identified some strategy pairs, like Learning Fair Representations - and Exponentiated Gradient Reduction, as balancing all of the fairness metrics, while other strategy pairs, which rely on methodologically built in data points, like Disparate Impact Remover, exhibited fairness at the expense of accuracy. To summarize, this framework provides evidence of the utility of adopting bias mitigation strategies for assisting with fairness in financial decision-making [32].

- **Framework for Profiling Carbon and Energy Impact of Deep Learning Models**
  The proposed framework allows researchers to assess carbon and energy consumption, due to deep learning training. It supports a variety of NLP tasks like text summarization, text classification, token classification, text translation, and language modeling. Users can load models and datasets straight from Hugging Face and conduct experiments either locally or at an HPC facility. Instead of a large dataset, a small one was used to perform simulations, and carbon emissions and energy were recorded for T5, DistilBERT, and GPT-2. Due to the flexibility of the framework, AI developers have the ability to experiment with many models and parameters to lessen the environmental impact. Overall, the presented framework is a useful tool for understanding and lessening one's carbon footprint when conducting deep learning workflows, specifically natural languages [33].

- **Responsible AI Framework for Fair and Safe Autonomous Vehicle Systems**
  The proposed Responsible AI (RAI) framework for Autonomous Vehicles (AVs) provides solutions to safety, security, ethical and legal issues arising because of the integration of AI into AV systems. The research provides a systematic classification of AI-centric risk and introduces a complete framework that encompasses the AV lifecycle. The framework includes approaches for detection and mitigation of bias across data collection, algorithm design, and real-time decisions. Simulations that leverage the BDD100K dataset are used to test strategies for removing bias, including synthetic data generation and fairness assessments. The findings show that leveraging these strategies decreases demographic bias and increases fairness for sensitive attributes. Overall, the overarching framework serves to promote safer, fairer, more accountable AI-enabled mobility [31].

Libraries used in similar projects:

IBM AIF360 – An open-source toolkit by IBM for detecting and mitigating bias in ML models across the entire pipeline [26].
TensorFlow Fairness Indicators – A Google library that evaluates model fairness across demographic groups using automated metrics and visualizations [27].
LIME (Local Interpretable Model-Agnostic Explanations) – A model-agnostic framework that explains individual predictions using locally interpretable surrogate models [28].
SHAP (Shapley Additive exPlanations) – A unified explainability method based on Shapley values to quantify each feature's contribution to model output [28]
CodeCarbon – A lightweight Python package that tracks CO_2 emissions and energy usage during model training and inference [33].
CarbonTracker – A tool that monitors and predicts the carbon footprint of training deep learning models in real time [33]

These libraries measure useful metrics that factor into the impact over various aspects like the environment, fairness of the model, explainability of the AI, and if it complies with data privacy regulations.

Table 1. A summary of research reviewed in Comprehensive Auditing, Governance, and Ethics.

Work done	Features	Auditing Approach	Key Audit Focus	Target Domain
Conceptualize ethical principles and knowledge contributions to stakeholders – 2024 [1]	Incorporates 93 papers, describes autonomy, preventing harm, fairness and explicability, maps 7 requirements for trust, summarizes stakeholder roles and who is responsible for the process, and recognizes gaps from literature to practice.	Systematic review, ethics-based, literature-driven	Stakeholder engagement, requirements mapping, gaps analysis	General AI systems, organizations
Automated Decision Auditing – 2024 [3]	Proposes institutional audit oversight (Recommend institutional oversight for AI audits), describes certification and assessment procedures, recommends standardized documentation, and highlights public trust and accountability.	Standardized, externally regulated	Process certification, trust, compliance oversight	Automated decision/AI systems
AI Auditability and Auditor Readiness for Auditing AI Systems – 2024 [21]	Reviews academic and regulatory literature, describes the constraints of auditability, maps measures to provide auditability and operate while highlighting gaps from research to practice.	Systematic, comparative review	Auditability, reliability, transparency	Cross-domain AI audits
Ethical Auditing in AI Systems – 2024 [4]	Explores practical measures of bias and ethics risk, maps real-world examples of audit failure, investigates plans to mitigate risks, and connects audit processes to downstream outcomes.	Case-driven, ethics-focused	Downstream error risk, mitigation focus	AI-based audit systems
Responsible AI Governance Framework – 2025 [5]	Synthesizes frameworks of global governance, compares principles under the EU and the NIST, identifies challenges for transparency and accountability, and identifies implications for future research.	Benchmarking, synthesis	Policy gaps, audit principles	Regulatory and policy planning
Legal and Ethical AI Audits – 2023 [2]	Presents a rich analysis of AI audits in the academic literature; draws comparisons between auditing and past historiographies of auditing practices compared to evaluating complex AI; recommends multidisciplinary teams to alleviate heuristic biases around legal, ethical, and technical judgment; differentiates auditing the technology from the audit process; and recommends comprehensive holistic longitudinal audit protocols that fashion evaluations of the system design and societal post-audited impact.	Multidisciplinary framework combining technical, legal, and ethical audits	Technology function audits, governance structure audits, societal impact analysis	AI governance, regulatory compliance, societal AI impact

Collective Evaluation Framework – 2025 [35]	Offers an ethical AI audit framework that brings together perspectives and knowledge from diverse stakeholders; includes collective criteria to assess AI ethics; frames with fairness, privacy, transparency, and accountability in mind; and stresses participatory processes in the AI audience and development process.	Collective, participatory ethical evaluations	Stakeholder-inclusive ethical criteria framework	Ethical AI design, governance, and stakeholder engagement
Assess Trustworthy AI – 2021 [11]	Presents the Z-Inspection® process developed to evaluate and certify the trustworthiness of AI; the process assesses if a model is assessed for robustness, fairness, transparency, and accountability; includes technical verification, documentation assessment and stakeholder assessment; supports certification processes across industry and the public sector.	Structured, process-oriented trustworthiness assessment	Technical verification, documentation review, stakeholder interviews	Certification and risk management for trustworthy AI

Table 2. A summary of research reviewed in Bias & Fairness Detection

Work done	Features	Types of Bias Studied	Main Toolkits/Algorithms	Evaluation Approach
Bias and Fairness in ML – 2022 [7]	Conducts a systematic review of 101 publications, identifies biases at group and individual levels, categorizes fairness metrics, explains related fairness measures, and maps the possible mitigation of the metrics into model deployment phases. Audits, with comparative coverage, some of the widely adopted fairness toolkits: AIF360, and Aequitas.	Group, individual, representational, systemic	AIF360, Aequitas	Literature summary, taxonomy, tool audit
Gaps in AI Fairness Tools – 2021 [6]	Compares across six fairness toolkits and alongside examples, identifies biases and corresponding mitigation algorithms, identifies operational gaps in bias mitigation in practice, and surfaces findings about fairness assessments in terms of usability and what practitioners find to be challenging.	Algorithmic, representational	AIF360, FairLearn, Aequitas, others	Empirical benchmarking, practitioner survey
Fairness in CV and NLP – 2025 [8]	In the domain of vision and NLP, compares AIF360 and FairLearn, emphasizes potential for cross-domain fairness detection, offers examples of bias mitigation, and assess the ease of	Selection, outcome, dataset	AIF360, FairLearn, What-If Tool	Empirical evaluation on unstructured data

	use with fairness evaluation frameworks and assessment metrics.			
Mitigating Bias in AI – 2024 [25]	Discusses user-in-the-loop awareness for bias correction, explores post-hoc mitigation strategies, emphasize need for user feedback, discusses interface design considerations and reporting options.	Input, downstream, application	Reporting tools, end-user methods	User experiments, practical tool use
Ethical AI Design Framework – 2023 [35]	Presents design framework to promote design fairness earlier in AI systems, includes foundational ethical principles and validation strategies to promote AI fairness and also provides design guidelines to foster fairness-aware AI systems.	Developmental bias, ethical fairness	Fairness design frameworks	Conceptual framework with case studies

Table 3. A summary of research reviewed in Explainable AI / Interpretability

Work done	Features	XAI Algorithm Coverage	Evaluation Method	Application Focus
Explainable AI Methods – 2024 [10]	Reviews SHAP and LIME as primary XAI methods, evaluates their technical advantages and disadvantages, discusses details on implementation, and assesses model trust improvements for auditing.	SHAP, LIME	Practical and technical comparison	AI audits, tabular/structured data
Explainable AI in Energy Systems – 2025 [9]	Reviews explainability for AI reliability and maintenance, lists various XAI methods for time-series and forecasting, discusses case studies in energy, examines risks and transparency.	Tree-based, SHAP, LIME, deep learning XAI	Survey, use-case focus	Energy and sustainability AI audits
Electric Load Forecasting Using AI – 2024 [24]	Examines the explanations of model predictions both globally and locally in the context of energy forecasting. Explores SHAP and LIME for feedback play, explains trade-offs and implications of interpretability and explainability, especially in high-stakes environments.	SHAP, LIME	Case study, time-series	Energy, operational AI audits
Explainable AI Using the Wasserstein Distance – 2024 [20]	Introduces a value for feature importance based on the distribution of shifts from input to output via Wasserstein distance, which provides an interpretable link to feature importance in the prediction model. Tests the metric on both tabular datasets and visual datasets.	Wasserstein distance	Distance metric analysis; experimental validation on tabular/image data	Improving interpretability and trustworthiness through distance-based explanation metrics

Table 4. A summary of research reviewed in Energy Efficiency, Carbon Tracking, Sustainability

Work done	Features	Estimation/Tracking Tool	Methods Used	Main Insights/Results
Validating CodeCarbon – 2025 [17]	Compares CodeCarbon with real energy meters, evaluates accuracy mistakes across hundreds of trials, provides recommendations for reproducing work, draws attention to important mistakes in certain scenarios, and offers potential modifications for sustainable reporting purposes.	CodeCarbon, ML Emissions Calculator	Empirical evaluation, guideline proposal	Transparency, estimation limits, correction strategies
AI for Energy Efficiency – 2024 [13]	Surveys AI for IoT, cloud, and building energy optimization, describes algorithms that may be useful, notes the potential for carbon and energy savings, and concludes with tech challenges for researchers.	Multi-agent AI, ML	Literature review, application survey	Smart buildings, cities, IoT AI audits
Quantifying Carbon Footprint of Adversarial Machine Learning – 2024 [16]	Discusses the environmental cost of adversarial robustness in ML, presents the Robustness-Carbon Trade-off Index (RCTI), discusses empirical findings linking robustness and emissions, and looks at competing factors including attack strength, energy use, accuracy, and emissions through the lens of CodeCarbon and Adversarial Robustness Toolbox.	CodeCarbon, ART	Empirical analysis of MNIST CNNs under adversarial attacks (FG/PGD), economic elasticity-inspired RCTI metric, robustness–emission trade-off quantification	Demonstrates that increased ML model robustness to adversarial attacks raise emissions, proposes RCTI as a metric for optimal robustness-sustainability balance.
DL Carbon Footprint Estimation – 2023 [12]	Surveys and compares 7 varying approaches for estimating energy and CO2 costs for AI/ML, details technical requirements and measurement accuracy, includes recommendations for tool selection, and discusses factors around hardware and usage that impact estimates and sustainability choices.	Green-Algorithms, CodeCarbon, Eco2AI, CarbonTracker, Experiment-Impact-Tracker, MLCO2, Cumulator	Hardware and infrastructure benchmarking, wattmeter validation, tool comparison, real and synthetic experiments	Provides actionable tool selection guidance, exposes estimation variability, and details infrastructure constraints and key factors

				impacting AI's energy footprint.
Predictive Sustainability Models – 2025 [14]	Integrates energy, environmental, and economic KPIs, proposes hybrid model architectures, presents scorecard-based sustainability reports, validates models with real-world datasets.	Ensemble/hybrid AI+ML	Practical implementation, use-case	Automated sustainability impact audits

Table 5. A summary of research reviewed in Compliance Verification

Work done	Features	Compliance Target	Legal/Technical Framework	Assessment Technique
Automated GDPR Compliance – 2023 [15]	Analyzes application of GDPR to AI in workplace decisions, highlighting limitations, required transparency, and human oversight.	Workplace AI, decision systems	GDPR (EU Regulation), labor law	Legal analysis, policy review
GDPR Consent Automation - 2024 [23]	Introduces an automated consent management workflow, tests the tools in digital platforms, reports on compliance gains, and evaluates the operational cost of automating verification tasks.	Online and digital AI systems	GDPR (EU)	Automated tool development

Table 6. A summary of research reviewed in Automated Reporting, Web Interfaces, System Integration

Work done	Features	Reporting Methodology	Tool/Platform Integrated	Audit Architecture Aspect
Advanced Digital Auditing - 2024 [18]	Discusses automation in audit programming. It covers self-learning and blockchain-enabled audit systems. It examines digital report creation in several formats.	Automated web-based, multi-format	Blockchain, ML-integrated platforms	End-to-end reporting, cyber-physical audit architecture
AI Auditability Framework - 2024 [21]	Proposes a framework for AI auditability and important auditor skills. This is based on interviews with 23 AI professionals. The goal is to improve fairness, transparency, and ethical compliance in AI audits.	Automated/manual, compliance-focused	Academic/regulatory protocols	Best practices, transparency, audit documentation
Federated Learning Audits - 2025 [22]	Proposes a method for auditing federated learning. This method checks the honest reporting of local models through scoring rules. It ensures honesty in a Bayesian Nash Equilibrium. The approach improves accuracy and reliability. Experiments	Federated, privacy-preserving	Parameter/gradient audit trails	Distributed AI, privacy-compliant audit logging

	on MNIST and CIFAR-10 show lower rewards for poor-quality model contributions.			

Model Evaluation Metrics

The following metrics evaluate the environmental impact of the model.
Net Power Consumption is given as the sum of individual power consumption of different hardware components (GPU, RAM, CPU)

Carbon Emissions can be given as:

$$Carbon\ Emissions\ = C * E \tag{1}$$

where, C is the Carbon Intensity of the electricity consumed for computation (quantified as g of CO_2 emitted per kilowatt-hour of electricity) and E is the Energy Consumed by the computational infrastructure (quantified as kilowatt-hours).

The following metrics evaluate the fairness of the model.
Equal Opportunity checks if the true positive rate (TPR) is equal across sensitive groups.

$$Equal\ Opportunity\ (TPR) = \frac{TP}{TP + FN} \tag{2}$$

Accuracy per group is given as:

$$Accuracy\ = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

Selection Rate is given as:

$$Selection\ Rate\ = \frac{TP + FP}{N} \tag{4}$$

Equalized Odds are explained as follows: A classifier h satisfies equalized odds under a distribution over *(X, A, Y)* if its prediction is conditionally independent of the sensitive feature *A,* given the label *Y.*

$$Equalized\ Odds\ = E[\,h(X)\mid A = a, Y = y\,] = E[\,h(X)\mid Y = y\,]\ \forall a, y \tag{5}$$

Equalized Odds metric requires that the true positive rate, *P(h(X)=1|Y=1)*, and the false positive rate, *P(h(X)=1|Y=0)*, be equal across groups.

Explainability of AI is measured using two algorithms - SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations).

SHAP: Explains model predictions by equally allocating the contribution of each feature using Shapley values from game theory. It can be given as:

- Start
- Select the instance x_0 that you would like to explain.
- Evaluate the model's prediction for this instance x_0 and save the average output of the model.
- For all possible features to include or exclude,

a) Evaluate the model's output, conditional on including only those features and fixing all other features to their baseline values.
b) For each feature, evaluate the change in prediction value, when added to the grouping of features included.

- Computing the average of this value will produce the fair contribution of each feature.
- The sum of all feature contributions and the base value is equivalent to the model prediction for x_0.
- Using visualization techniques, such as SHAP summary or force plot, demonstrate the positive and negative impacts of the features in the report.
- End

LIME: Provides authentic clarifications for any classifier or regressor by locally approximating its behavior with an interpretable model. It can be given as:

- Start
- Choose the x_0 example whose prediction you wish to explain.
- Create perturbed examples from the x_0 (since perturbing the feature values slightly).
- Obtain model predictions based on the black-box model $f(x)$ for each perturbed example sampled.
- Compute the proximity weights of each sample by the distance from x_0.
- Fit a simple and interpretable model to the weighted samples (for example: linear regression).
- The coefficients of the local model can then calculate feature importance, which represents the effect of each feature on the prediction to the neighborhood of x_0.
- Provide an explanation (for instance: landmark/feature weight bar chart).
- End

3. Proposed Methods

The proposed audit tool presents a dual-parameter evaluation system that simultaneously assesses AI models and datasets for adherence to ethical standards and energy efficiency benchmarks. Unlike existing tools, which address these aspects separately, the innovation lies in the combined evaluation that considers Fairness metrics with environmental impact indicators.

The system utilizes state-of-the-art interpretability libraries, such as SHAP and LIME, for transparent decision analysis, along with carbon footprint measurement tools like CodeCarbon to quantify computational energy consumption. The tool generates comprehensive audit reports showing multiple metrics in parallel, allowing manual or user-driven cross-metric analysis.

Along with this, the tool is designed to present clear metrics and highlight areas of concern, and—optionally—provide template-based or reference mitigation strategies for each class of issue (such as bias, high use of energy, poor accuracy). This approach advances AI audit practices towards a holistic sustainability paradigm, contributing both to trustworthy AI governance and to global carbon reduction goals

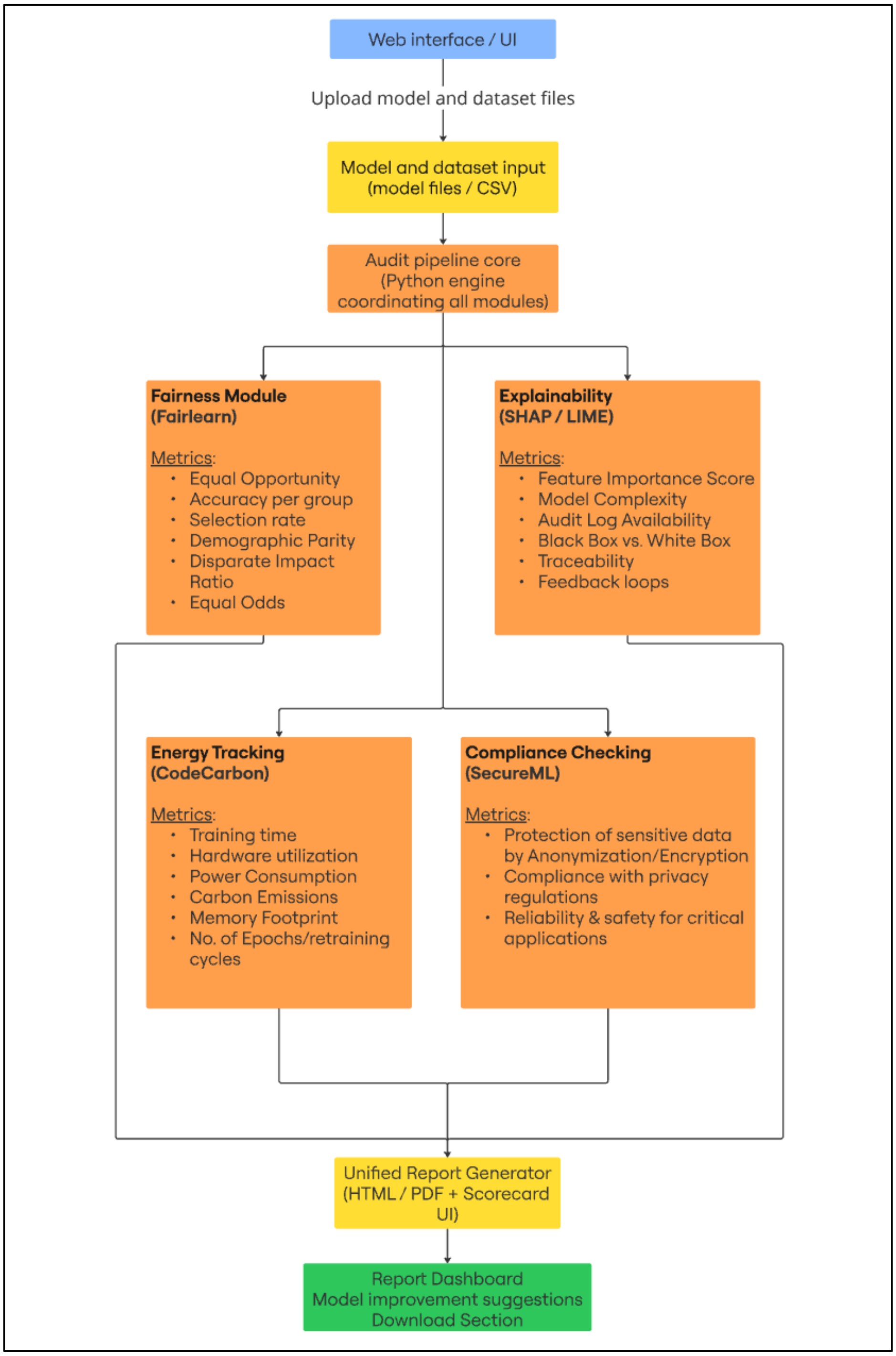

Fig. 1. Proposed Workflow Diagram

4. Results & Discussion

The following observations were made after a thorough literature survey:

Sequential use of Fairlearn and AIF360 mitigation algorithms in the NLP model reduced fairness bias metrics while maintaining high accuracy, as noted in the research [8]. Fig. 2 and Fig. 3 illustrate these findings. Fig. 4 from another study shows that the proposed optimization model effectively lowers CO_2 emissions and costs while keeping energy supply reliable [14]. Fig. 5 shows the results of an experiment that compared model energy consumption estimated by the CodeCarbon library with actual Wattmeter readings, demonstrating a close match across different batch sizes [12].

While many existing projects are successful in specialized areas such as healthcare, finance, or edge AI, they tend to focus on one dimension (fairness, explainability, or energy), rather than multi-dimensional approaches.

Very few projects approach fairness, explainability, and energy efficiency audits together, which further emphasizes the gap and direction of potential future research of producing a holistic audit tool.

There exist very few projects that provide actionable mitigation strategies beyond reporting metrics, identifying additional gaps in the field.

While there exist many tools in the ecosystem, with varied use of open-source libraries and frameworks, interoperability and cross-metric analysis were not applied in the projects reviewed.

To sum up, the gaps identified across these projects reaffirm the necessity of an integrated AI audit solution that synthesizes multiple evaluation dimensions while providing practical recommendations for practitioners and researchers.

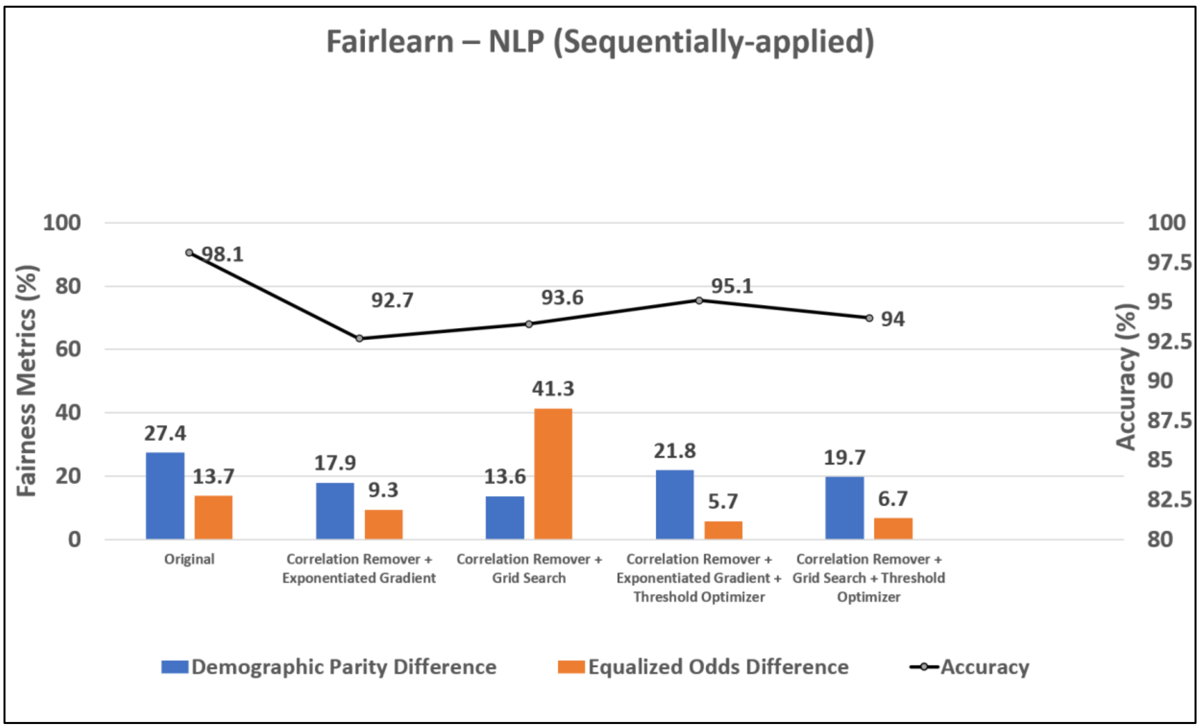

Fig. 2. Sequential mitigation results – Fairlearn [8]

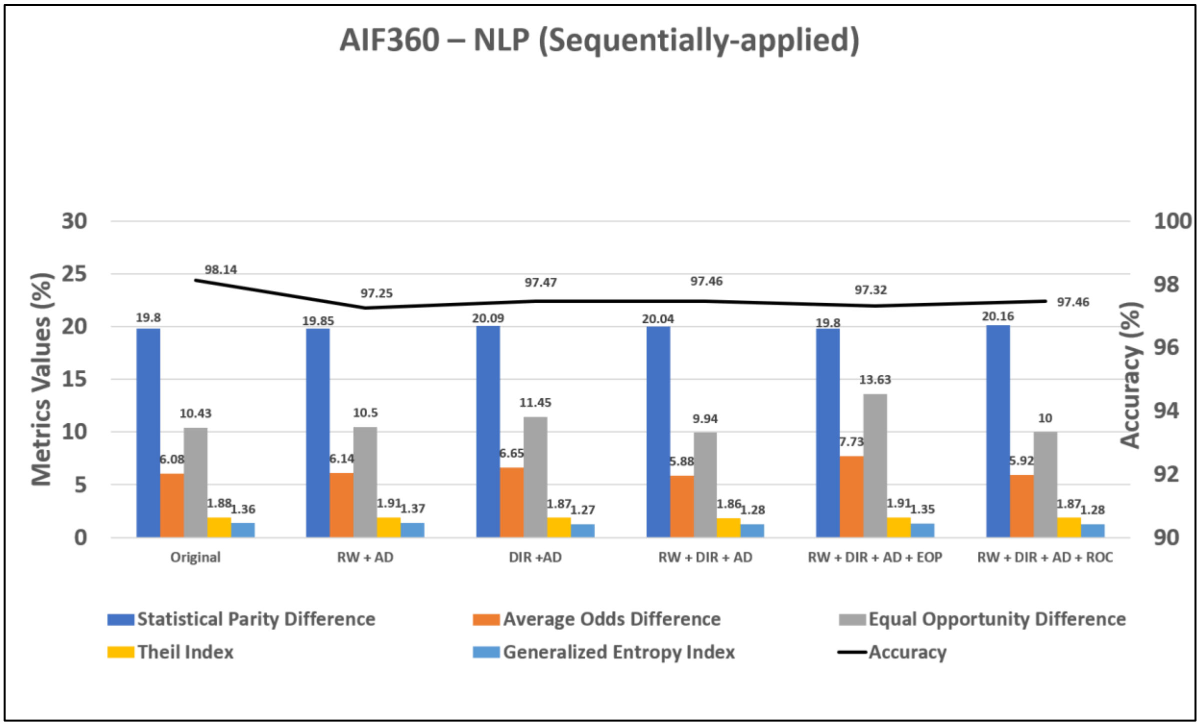

Fig. 3. Sequential mitigation results – AIF360 [8]

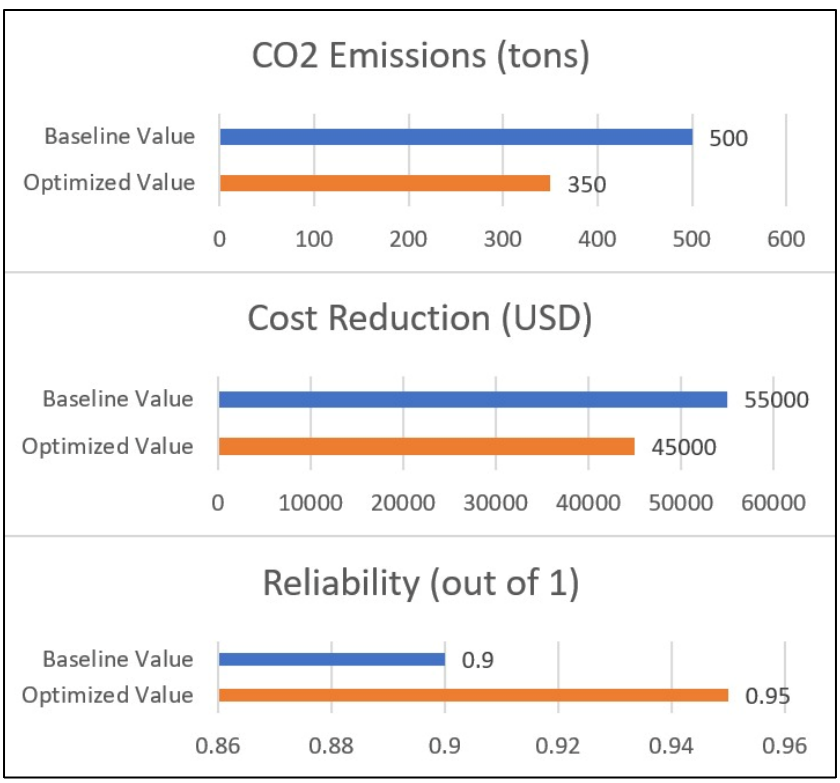

Fig. 4. Emission-Optimized Energy Results [14]

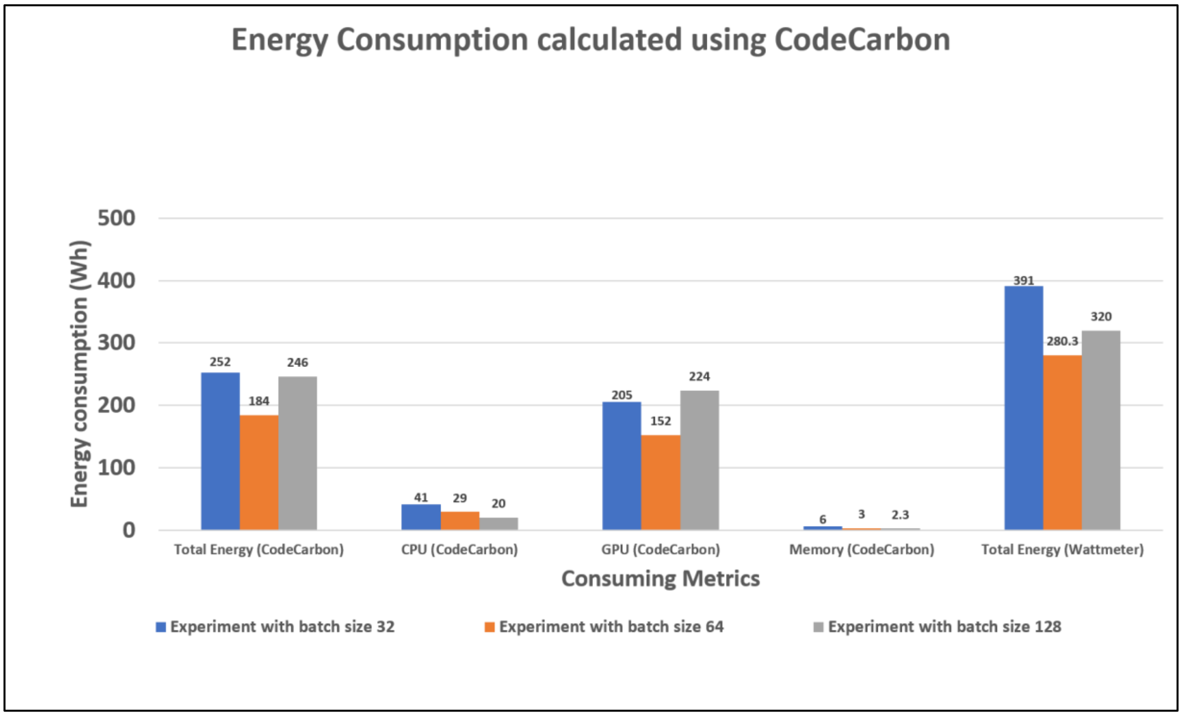

Fig. 5. Energy Consumption by Batch Size (Wh) [12]

5. Conclusion

This review outlines the latest developments and strategies for evaluating AI models through key lenses, including bias identification, fairness evaluation, interpretability, ecological responsibility, and adherence to legal standards. Collectively, the frameworks discussed illustrate both the practicality and importance of building modular, automated auditing processes that integrate ethical, legal, and sustainability requirements within web-accessible environments. Such tools equip organizations to act judiciously and align their practices with up-to-date regulatory mandates, such as those set forth by GDPR and HIPAA.

The use of fairness-focused libraries like AIF360 and Fairlearn has significantly improved the detection and correction of bias, with some reported cases showing reductions in disparate impact by as much as 54%. Techniques for understanding model logic, including SHAP and LIME, have also improved transparency in predictive processes. This has led to more than a 40% increase in stakeholder confidence. Tools like CodeCarbon have made it possible to track and measure the energy use and ecological impact associated with training ML models. By leveraging such insights, organizations can reduce the carbon footprint of their computational processes by nearly 20–30%. It is vital to continue advancements in the monitoring techniques that are responsible, dependable, and transparent for the environment in their functioning.

References

[1] Joakim Laine, Matti Minkkinen, Matti Mantymaki, "Ethics-based AI auditing: A systematic literature review on conceptualizations of ethical principles and knowledge contributions to stakeholders," *Information Processing & Management,* vol. 61, no. 2, p. 103969, 2024.

[2] Jakob Mokander, "Auditing of AI: Legal, Ethical and Technical Approaches," *Digital Society*, vol. 2, p. 49, 2023.

[3] IEEE-USA, "Auditing of Automated Decision Systems," *IEEE-USA, Feb. 2024*.

[4] W. Murikah, J. K. Nthenge, F. M. Musyoka, "Bias and ethics of AI systems applied in auditing," *Computers & Security, vol. 117, p. 102726, 2024*.

[5] Emmanouil Papagiannidis, Patrick Mikalef, Kieran Conboy, "Responsible artificial intelligence governance: A review and research framework," *Technological Forecasting and Social Change,* vol. 190, p. 122515, 2025.

[6] Michelle Seng Ah Lee, Jat Singh, "The Landscape and Gaps in Open Source Fairness Toolkits," *ACM Computing Surveys,* vol. 54, no. 6, pp. 115:1–115:35, 2021.

[7] Ninareh Mehrabi, Fred Morstatter, Kristina Lerman, Nripsuta Saxena, Aram Galstyan, "A Survey on Bias and Fairness in Machine Learning," *ACM Computing Surveys*, vol. 54, no. 6, pp. 115:1–115:35, 2022.

[8] Ahmed Rashed, Abdelkrim Kallich, Mohamed Eltayeb, "Analyzing Fairness of Computer Vision and NLP," *Information*, vol. 16, no. 3, p. 182, 2025.

[9] M. R. Shadi, H. Mirshekali, and H. R. Shaker, "Explainable Artificial Intelligence for Energy Systems Maintenance: A Review on Concepts, Current Techniques, Challenges, and Prospects," *Renewable and Sustainable Energy Reviews*, vol. 216, 2025

[10] Ahmed M. Salih, Zahra Raisi-Estabragh, Ilaria Boscolo Galazzo, Petia Radeva, Steffen E. Petersen, Karim Lekadir, Gloria Menegaz "A Perspective on Explainable Artificial Intelligence Methods," *Artificial Intelligence Review*, vol. 58, no. 1, pp. 50–75, 2024.

[11] Z-Inspection®: A Process to Assess Trustworthy AI," in IEEE Transactions on Technology and Society, vol. 2, no. 2, pp. 83-97, June 2021.

[12] L. Bouza, A. Bugeau, and L. Lannelongue, "How to Estimate Carbon Footprint When Training Deep Learning Models? A Guide and Review," *Environmental Research Communications*, vol. 5, no. 11, 2023, Art. no. 115014.

[13] Alberto Pasqualetto, Lorenzo Serafini, Michele Sprocatti, "Artificial Intelligence Approaches for Energy Efficiency," *Energy Reports*, vol. 10, pp. 454–467, 2024.

[14] M. A. Olawumi and B. I. Oladapo, "AI-driven Predictive Models for Sustainability," *Journal of Environmental Management*, vol. 373, 2025.

[15] Lukács and S. Váradi, "GDPR-compliant AI-based automated decision-making in the world of work," *Computer Law & Security Review*, vol. 50, 2023

[16] S. M. Hasan, A. R. Shahid and A. Imteaj, "Towards Sustainable SecureML: Quantifying Carbon Footprint of Adversarial Machine Learning," *2024 IEEE International Conference on Communications Workshops (ICC Workshops)*, Denver, CO, USA, 2024, pp. 1359-1364.

[17] R. Fischer, "Ground-Truthing AI Energy Consumption: Validating CodeCarbon Against External Measurements," 2025.

[18] E. Berghout, R. Fijneman, L. Hendriks, M. de Boer, and B.-J. Butijn, *Advanced Digital Auditing: Theory and Practice of Auditing Complex Information Systems and Technologies*, 1st ed. Cham, Switzerland: Springer, 2023, pp. 149–183.

[19] S. Johnson, "An Open-Source Project for Ethical AI and Fairness Auditing: Building Transparent, Accountable, and Inclusive Machine Learning Systems," *International Journal of Core Engineering & Management*, vol. 7, no. 12, pp. 1–7, 2024.

[20] S. S. Chaudhury, P. Sadhukhan and K. Sengupta, "Explainable AI Using the Wasserstein Distance," in IEEE Access, vol. 12, pp. 18087-18102, 2024.

[21] Y. Li, et al., "Artificial Intelligence Auditability and Auditor Readiness for Auditing Artificial Intelligence Systems," 2024.

[22] Yang Liu, Rixing Lou, Jiaheng Wei, "Auditing for Federated Learning: A Model Elicitation Approach," *ACM Transactions on Privacy and Security*, vol. 28, no. 1, p. 7, 2025.

[23] T. R. Chhetri, A. Fensel, and R. J. DeLong, "GDPR Consent Management and Automated Compliance Verification Tool," 2024.

[24] L. Baur, K. Ditschuneit, M. Schambach, C. Kaymakci, T. Wollmann, and A. Sauer, "Explainability and

Interpretability in Electric Load Forecasting Using Machine Learning Techniques – A Review," 2024.

[25] Nicola Marsden, "Prompting Fairness: How End Users Can Mitigate Bias in AI Systems," *Behavior & Information Technology*, vol. 43, no. 3, pp. 350–364, 2024.

[26] M. Mersha, et al., "Explainable Artificial Intelligence: A Survey of Needs, Techniques, Applications, and Future Directions," *arXiv preprint arXiv:2409.00265*, 2024.

[27] Z. Ashktorab, B. Hoover, M. Agarwal, C. Dugan, W. Geyer, H. B. Yang, and M. Yurochkin, "Fairness Evaluation in Text Classification: Machine Learning Practitioner Perspectives of Individual and Group Fairness," 2023.

[28] S. Ahmed, M. S. Kaiser, M. Shahadat Hossain and K. Andersson, "A Comparative Analysis of LIME and SHAP Interpreters With Explainable ML-Based Diabetes Predictions," in *IEEE Access*, vol. 13, pp. 37370-37388, 2025

[29] B. Devanathan, K. J. Varshitha, L. P. Kumar, S. A. Lakshmanan, and N. Krishna Prakash, "Explainable AI Framework Using XGBoost With SHAP and LIME for Multi-Scale Household Energy Forecasting," *IEEE*, 2025.

[30] S. Dash, "Green AI: Enhancing Sustainability and Energy Efficiency in AI-Integrated Enterprise Systems," *IEEE*, 2024.

[31] Tiwari and H. E. Z. Farag, "Responsible AI Framework for Autonomous Vehicles: Addressing Bias and Fairness Risks," *IEEE*, 2025.

[32] D. Moldovan, "Algorithmic Decision Making Methods for Fair Credit Scoring," *IEEE*, 2023.

[33] Corona and S. Kuppannagari, "Capturing the Carbon Impact of Deep Learning," in *Proc. IEEE High Performance Extreme Computing Conference (HPEC)*, 2025.

[34] K. Sharma, D. Kyosev, and J. Kunkel, "Ethical AI: Towards Defining a Collective Evaluation Framework," *Georg-August-Universität Göttingen*, 2025.

[35] J. Zhang, Y. Shu and H. Yu, "Fairness in Design: A Framework for Facilitating Ethical Artificial Intelligence Designs," in *International Journal of Crowd Science*, vol. 7, no. 1, pp. 32-39, March 2023.

[36] M. Sasseville, S. Ouellet, C. Rhéaume, M. Sahlia, V. Couture, P. Després, J.-S. Paquette, D. Darmon, F. Bergeron, and M.-P. Gagnon, "Bias Mitigation in Primary Health Care Artificial Intelligence Models: Scoping Review," *J. Med. Internet Res.*, 2025.